

AIR UNIVERSITY, AEROSPACE AND AVIATION CAMPUS, KAMRA

NAME	**EIMAN FATIMA**
REG ID	**245237**
SUBMITTED TO	**MENAHIL NASEER**
LAB	**5**
DATE	**10/12/2026**

Task 1: Named Routes — Mini Recipe App

Apply named routes to build a 3-screen recipe app: a RecipeListScreen, a RecipeDetailsScreen, and an AboutScreen.

• Register all three screens in MaterialApp's routes table using string route names ('/', '/details', '/about').

• Tapping a recipe in the list must navigate to '/details' using Navigator.pushNamed, passing the recipe name as an argument.

• RecipeDetailsScreen must read and display the passed argument using ModalRoute.of(context)!.settings.arguments.

Deliverable: source code plus screenshots showing navigation to the details screen with the correct recipe name displayed.

CODE:



```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      initialRoute: '/',
      routes: {
        '/': (context) => const HomeScreen(),
        '/about': (context) => const AboutScreen(),
      },
    );
  }
}

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Home')),
      body: Center(
        child: ElevatedButton(
          onPressed: () {
```

```
            onPressed: () {
              Navigator.pushNamed(context, '/about');
            },
            child: const Text('About Me'),
          ), ElevatedButton
        ), Center
      ); Scaffold
    }
}

class AboutScreen extends StatelessWidget {
  const AboutScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('About')),
      body: const Center(
        child: Text('Welcome to my app!'),
      ), Center
    ); Scaffold
  }
}
```

OUTPUT:

```
import 'package:flutter/material.dart';
void main() {
  runApp(const RecipeApp());
}
class RecipeApp extends StatelessWidget {
  const RecipeApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Recipe App',

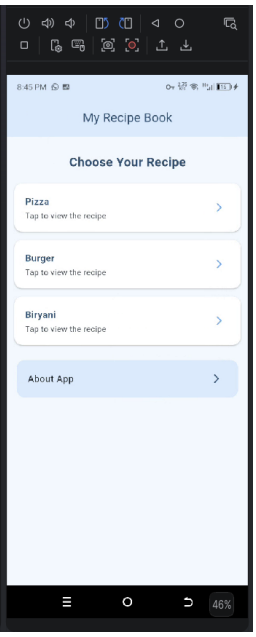
      theme: ThemeData(
        scaffoldBackgroundColor: const Color(0xFFF5F9FF),

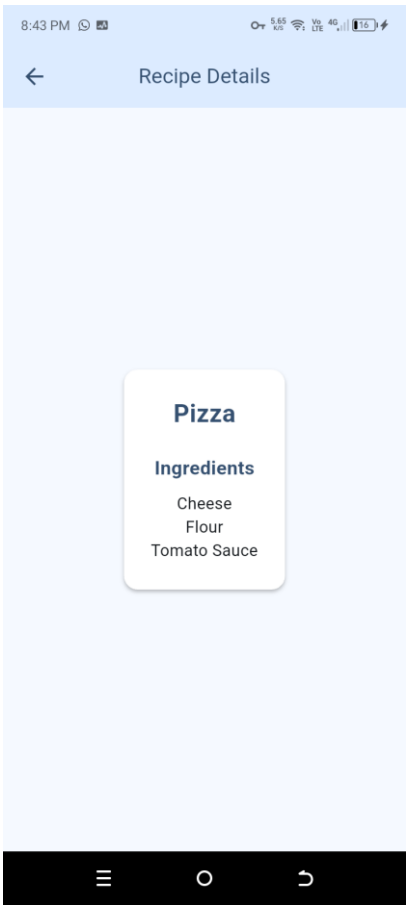
        colorScheme: ColorScheme.fromSeed(
          seedColor: const Color(0xFFDCEBFF),
          brightness: Brightness.light,
        ),

        appBarTheme: const AppBarTheme(
          backgroundColor: Color(0xFFDCEBFF),
          foregroundColor: Color(0xFF355070),
          elevation: 0,
        ),
      ),

      initialRoute: '/',

      routes: {
        '/': (context) => const RecipeListScreen(),
```







Task 2: Hero Animation — Product Gallery

Apply the Hero widget to build a 2-screen product gallery: a grid/list of product thumbnails, and a ProductDetailsScreen showing

the same image enlarged.

• Use matching Hero tags (e.g., the product's unique ID) on both the thumbnail and the enlarged image.

• Tapping a thumbnail must navigate to the details screen with a visibly smooth shared-element transition.

• Apply this to at least three different products, each with its own unique Hero tag.

Deliverable: source code plus a short screen recording showing the Hero transition for at least one product.

CODE:

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      initialRoute: '/',
      routes: {
        '/': (context) => const HomeScreen(),
        '/details': (context) => const DetailsScreen(),
      },
    );
  }
}

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('My Skills')),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
```

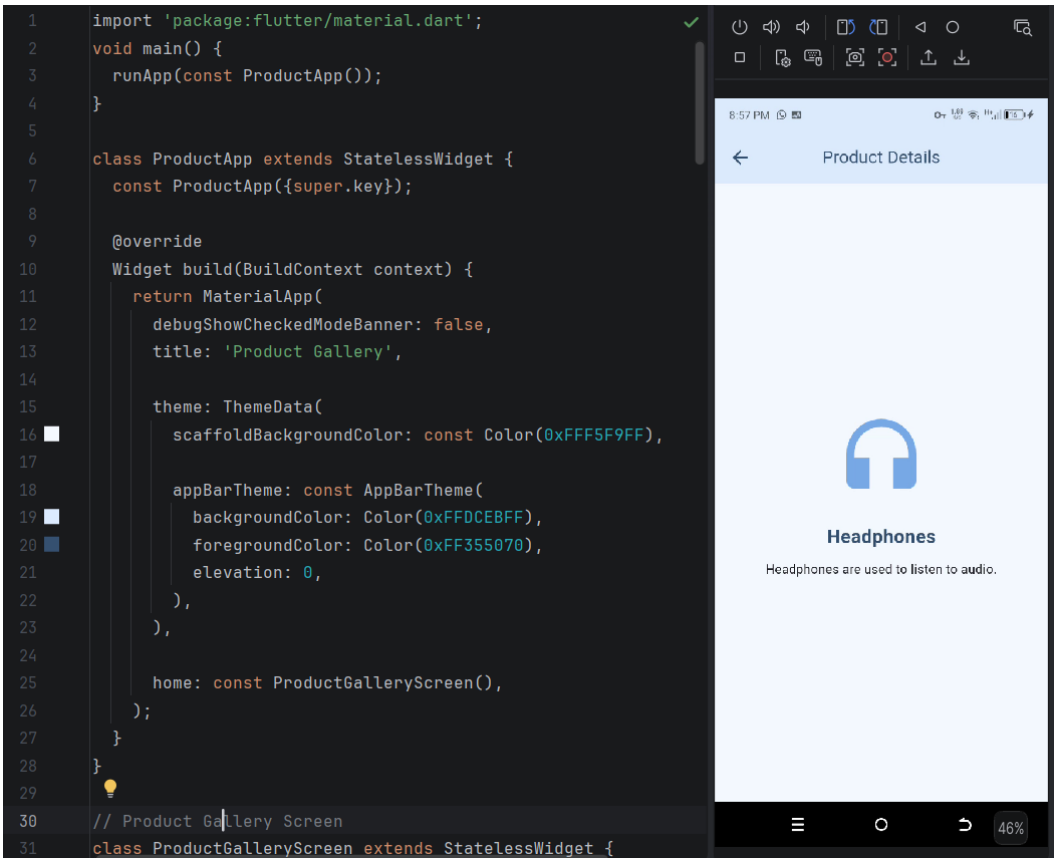
```
            children: [
              ElevatedButton(
                onPressed: () {
                  Navigator.pushNamed(
                    context,
                    '/details',
                    arguments: 'C++ Programming',
                  );
                },
                child: const Text('C++'),
              ), ElevatedButton
              ElevatedButton(
                onPressed: () {
                  Navigator.pushNamed(
                    context,
                    '/details',
                    arguments: 'Python Programming',
                  );
                },
                child: const Text('Python'),
              ), ElevatedButton
            ],
          ), Column
        ), Center
      ); Scaffold
    }
  }

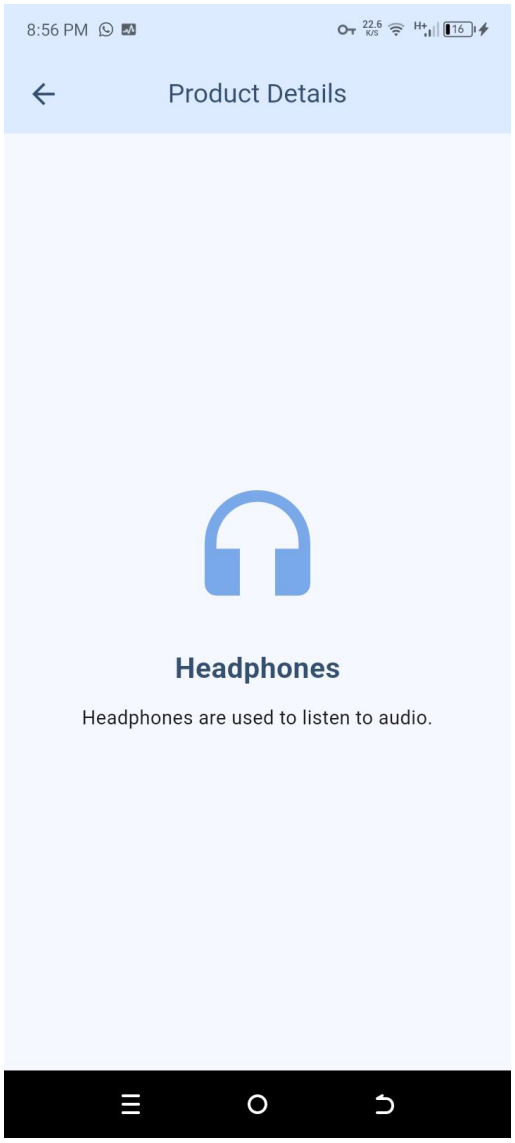
  class DetailsScreen extends StatelessWidget {
    const DetailsScreen({super.key});

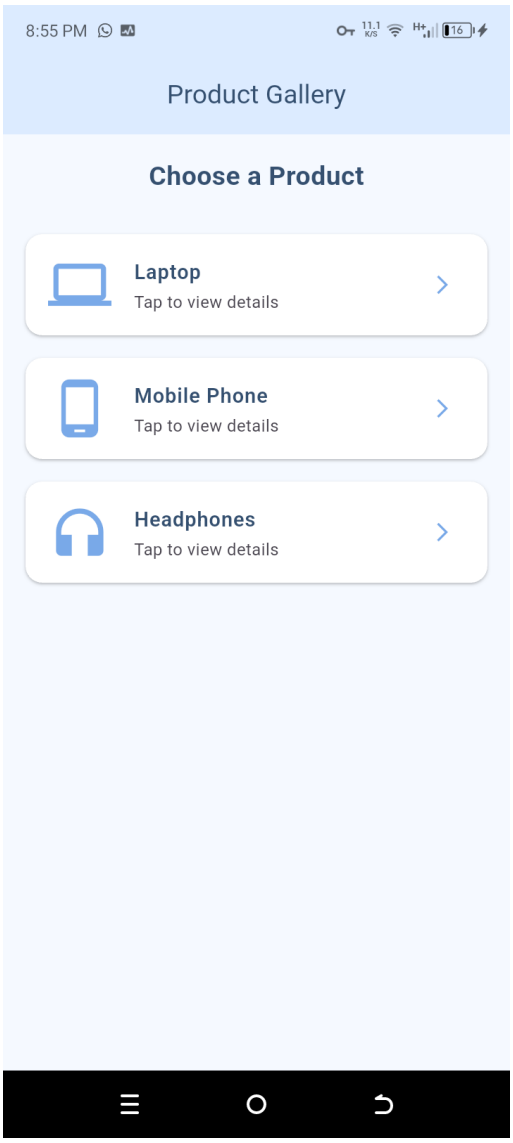
  @override
  Widget build(BuildContext context) {
    final skill =
    ModalRoute.of(context)!.settings.arguments as String;

    return Scaffold(
      appBar: AppBar(title: const Text('Skill Details')),
      body: Center(
        child: Text(
          skill,
          style: const TextStyle(fontSize: 24),
        ), Text
      ), Center
    ); Scaffold
  }
}
```

OUTPUT:


import 'package:flutter/material.dart';
void main() {
runApp(const ProductApp());
}
class ProductApp extends StatelessWidget {
const ProductApp({super.key});
@override
Widget build(BuildContext context) {
return MaterialApp(
debugShowCheckedModeBanner: false,
title: 'Product Gallery',
theme: ThemeData(
scaffoldBackgroundColor: const Color(0xFFF5F9FF),
appBarTheme: const AppBarTheme(
backgroundColor: Color(0xFFDCEBFF),
foregroundColor: Color(0xFF355070),
elevation: 0,
),
),
home: const ProductGalleryScreen(),
);
}
}
// Product Gallery Screen
class ProductGalleryScreen extends StatelessWidget {
8:57 PM
Product Details
Headphones
Headphones are used to listen to audio.
46%





Task 3: WebView — Embedded Help Center

Design and develop a HelpScreen that uses the WebView widget to embed a real website of your choice (e.g., your university's website or Flutter's documentation).

• Add the webview_flutter package and configure the required Android/iOS permissions.

• Include a WebViewController with at least two controls in the AppBar: a reload button and a back button (goBack() if

canGoBack() is true).

• Show a loading indicator while the page is loading (NavigationDelegate's onPageStarted/onPageFinished).

Deliverable: source code plus screenshots showing the embedded page loaded, and the reload/back controls working.

CODE:



```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const HomeScreen(),
    );
  }
}

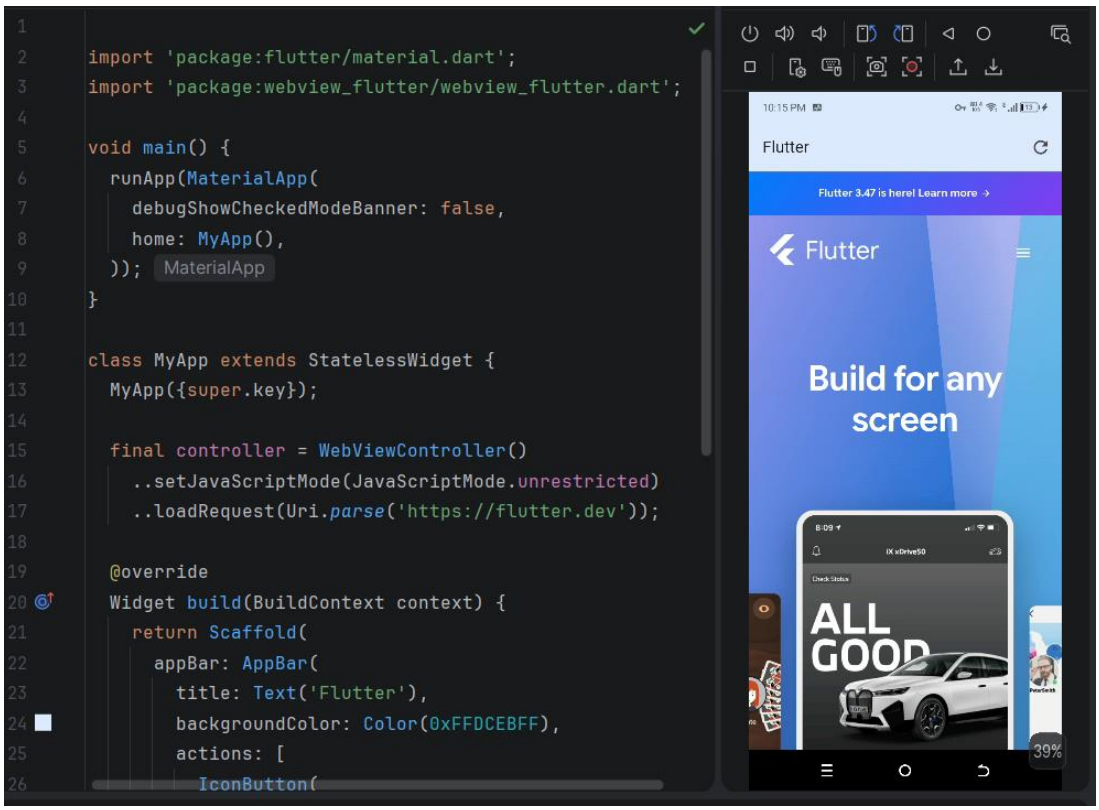
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

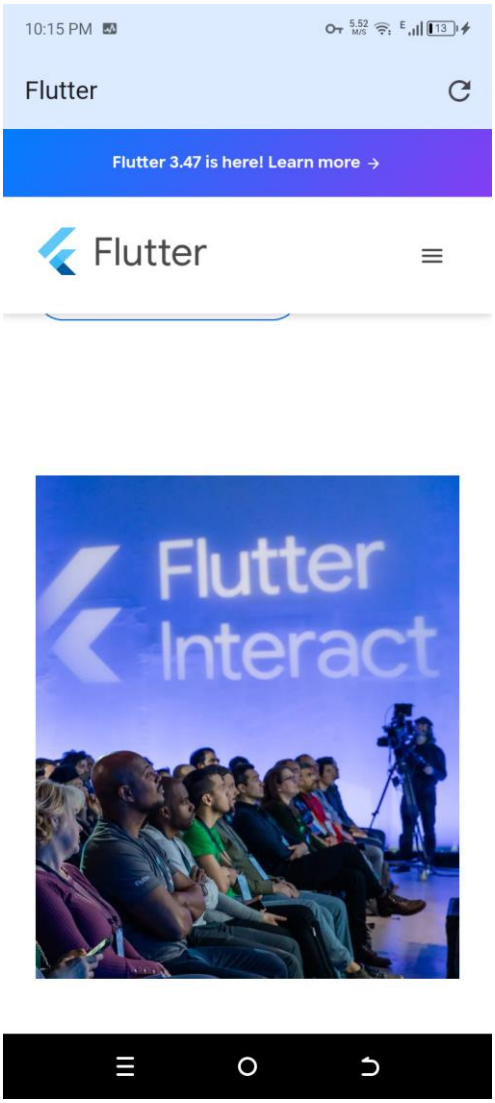
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Hero Animation')),
      body: Center(
        child: GestureDetector(
          onTap: () {
            Navigator.push(
              context,
              MaterialPageRoute(
                builder: (context) => const DetailsScreen(),
```

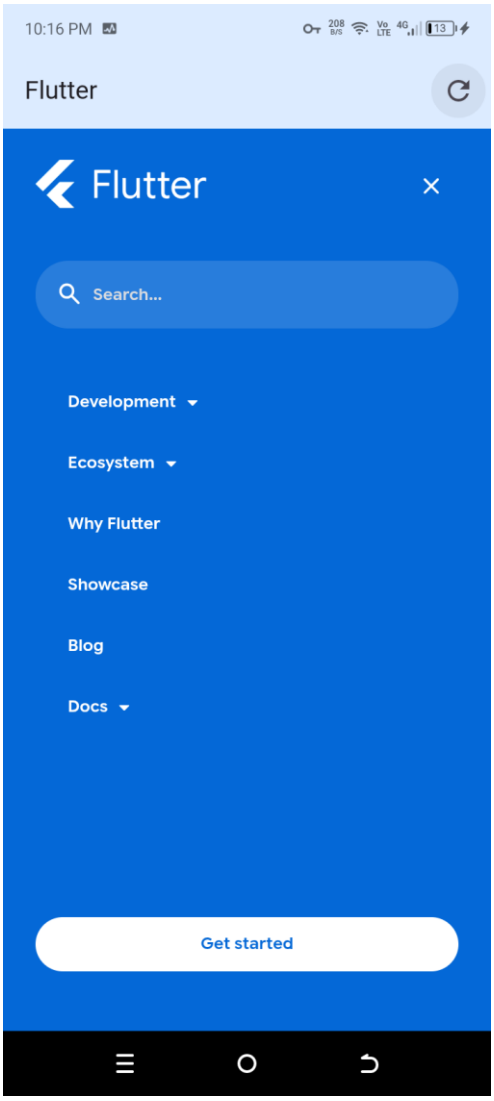
```
32                    ), MaterialPageRoute
33                  );
34                },
35                child: const Hero(
36                  tag: 'profile',
37                  child: CircleAvatar(
38                    radius: 50,
39                    backgroundColor: Color(0xFFDCEBFF),
40                    child: Icon(
41                      Icons.person,
42                      size: 60,
43                      color: Colors.blue,
44                    ), Icon
45                  ), CircleAvatar
46                ), Hero
47              ), GestureDetector
48            ), Center
49          ); Scaffold
50        }
51      }
52
53      class DetailsScreen extends StatelessWidget {
54        const DetailsScreen({super.key});
55
56        @override
57        Widget build(BuildContext context) {
58          return Scaffold(
59            appBar: AppBar(title: const Text('Profile')),
60            body: const Center(
61              child: Hero(
62                tag: 'profile',
```

```
            child: CircleAvatar(
              radius: 120,
              backgroundColor: Color(0xFFDCEBFF),
              child: Icon(
                Icons.person,
                size: 150,
                color: Colors.blue,
              ), Icon
            ), CircleAvatar
          ), Hero
        ), Center
      ); Scaffold
    }
  }
```

OUTPUT:







Task 4: Design Your Own App (Open-Ended)

Design and develop one small multi-screen app of your own choosing that applies all four concepts from this lab together

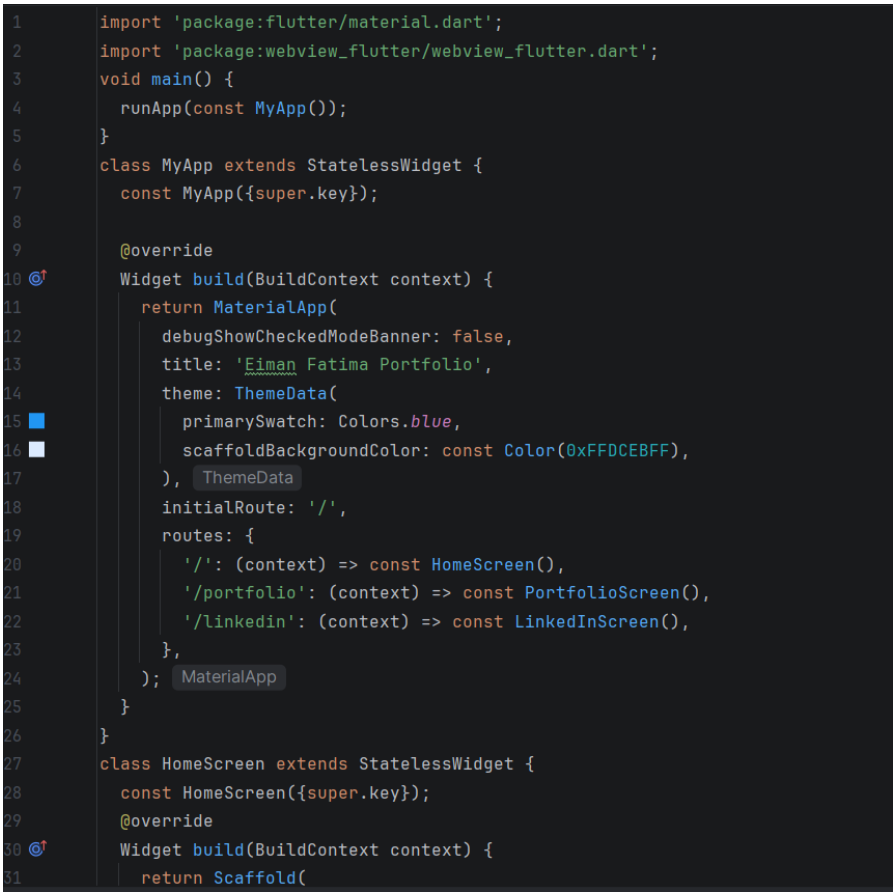
named routes, a Hero animation between two screens, and a WebView screen embedded somewhere in the app (e.g., a "Help", "About", or "Terms" page).

• You choose the app concept (e.g., event booking, campus info app, portfolio app, mini shop).

• Justify in a few sentences which screens use named routes, what the Hero animation connects, and why the WebView content belongs where you placed it.

Deliverable: working source code, plus a short written justification of your design choices. There is no single correct layout for this task — you are assessed on sound reasoning and correct application of the widgets covered in this lab.

CODE:



```dart
import 'package:flutter/material.dart';
import 'package:webview_flutter/webview_flutter.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Eiman Fatima Portfolio',
      theme: ThemeData(
        primarySwatch: Colors.blue,
        scaffoldBackgroundColor: const Color(0xFFDCEBFF),
      ), ThemeData
      initialRoute: '/',
      routes: {
        '/': (context) => const HomeScreen(),
        '/portfolio': (context) => const PortfolioScreen(),
        '/linkedin': (context) => const LinkedInScreen(),
      },
    ); MaterialApp
  }
}
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

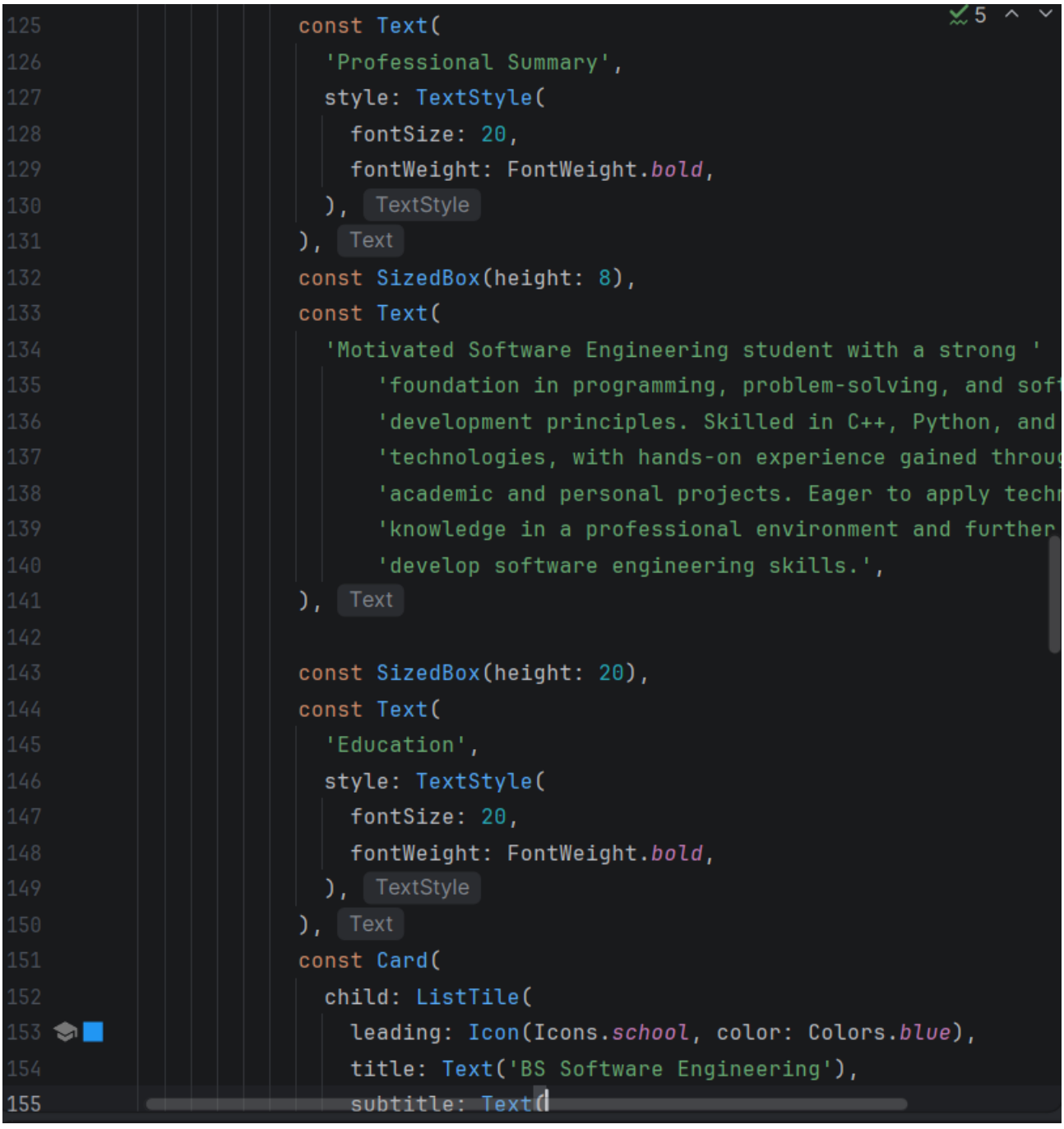
```
32          appBar: AppBar(
33            title: const Text('My Portfolio'),
34            backgroundColor: const Color(0xFFDCEBFF),
35          ), AppBar
36          body: Center(
37            child: Column(
38              mainAxisAlignment: MainAxisAlignment.center,
39              children: [
40                Hero(
41                  tag: 'profile',
42                  child: CircleAvatar(
43                    radius: 55,
44                    backgroundColor: Colors.blue,
45                    child: const Icon(
46                      Icons.person,
47                      size: 65,
48                      color: Colors.white,
49                    ), Icon
50                  ), CircleAvatar
51                ), Hero
52                const SizedBox(height: 20),
53                const Text(
54                  'EIMAN FATIMA',
55                  style: TextStyle(
56                    fontSize: 26,
57                    fontWeight: FontWeight.bold,
58                  ), TextStyle
59                ), Text
60                const SizedBox(height: 8),
61                const Text('Software Engineering Student'),
62                const Text('Aspiring Software Developer'),
```

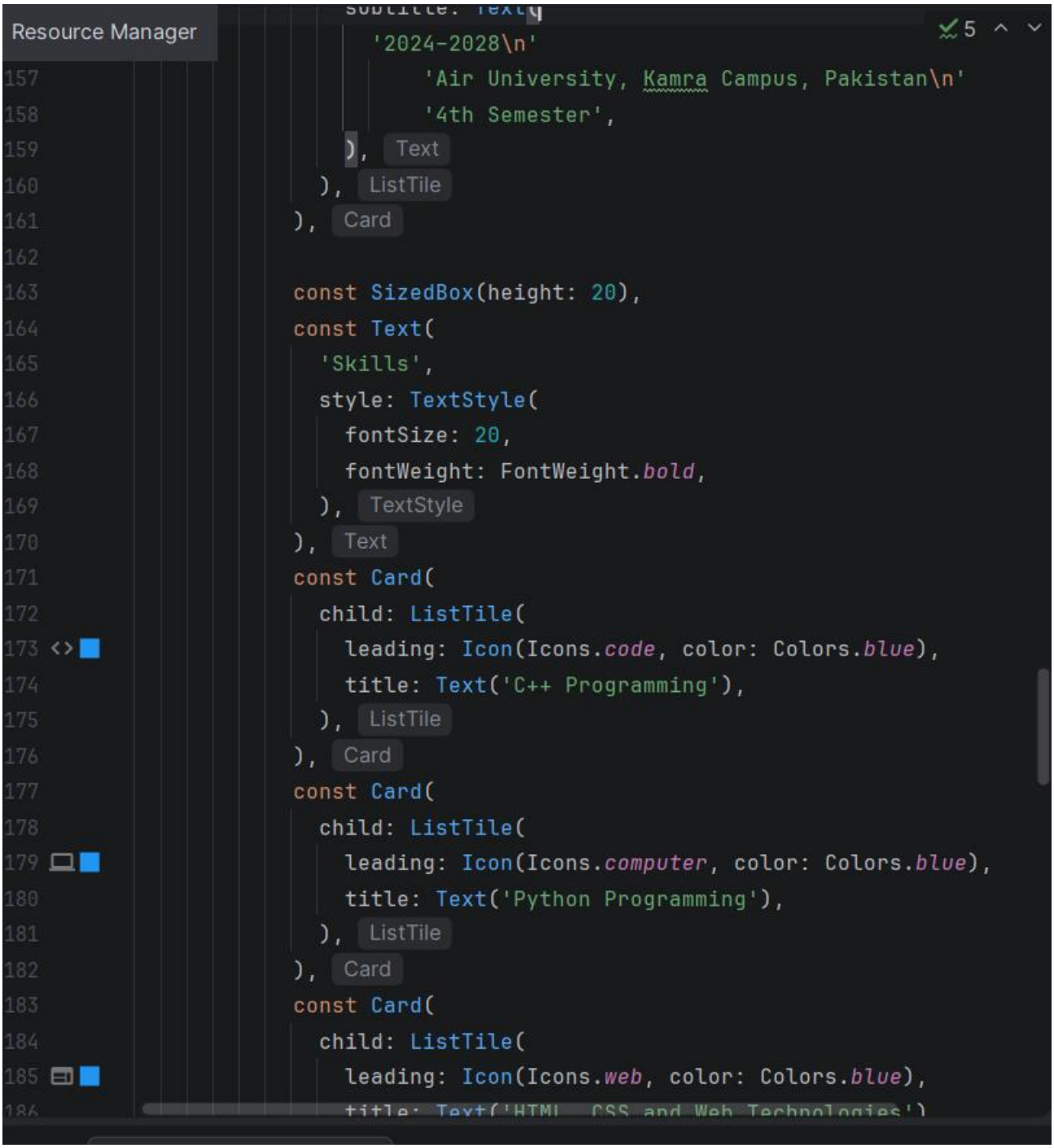
```dart
                const SizedBox(height: 25),
                ElevatedButton(
                  onPressed: () {
                    Navigator.pushNamed(context, '/portfolio');
                  },
                  child: const Text('View My Profile'),
                ), ElevatedButton
                ElevatedButton(
                  onPressed: () {
                    Navigator.pushNamed(context, '/linkedin');
                  },
                  child: const Text('My LinkedIn'),
                ), ElevatedButton
              ],
            ), Column
          ), Center
        ); Scaffold
      }
    }

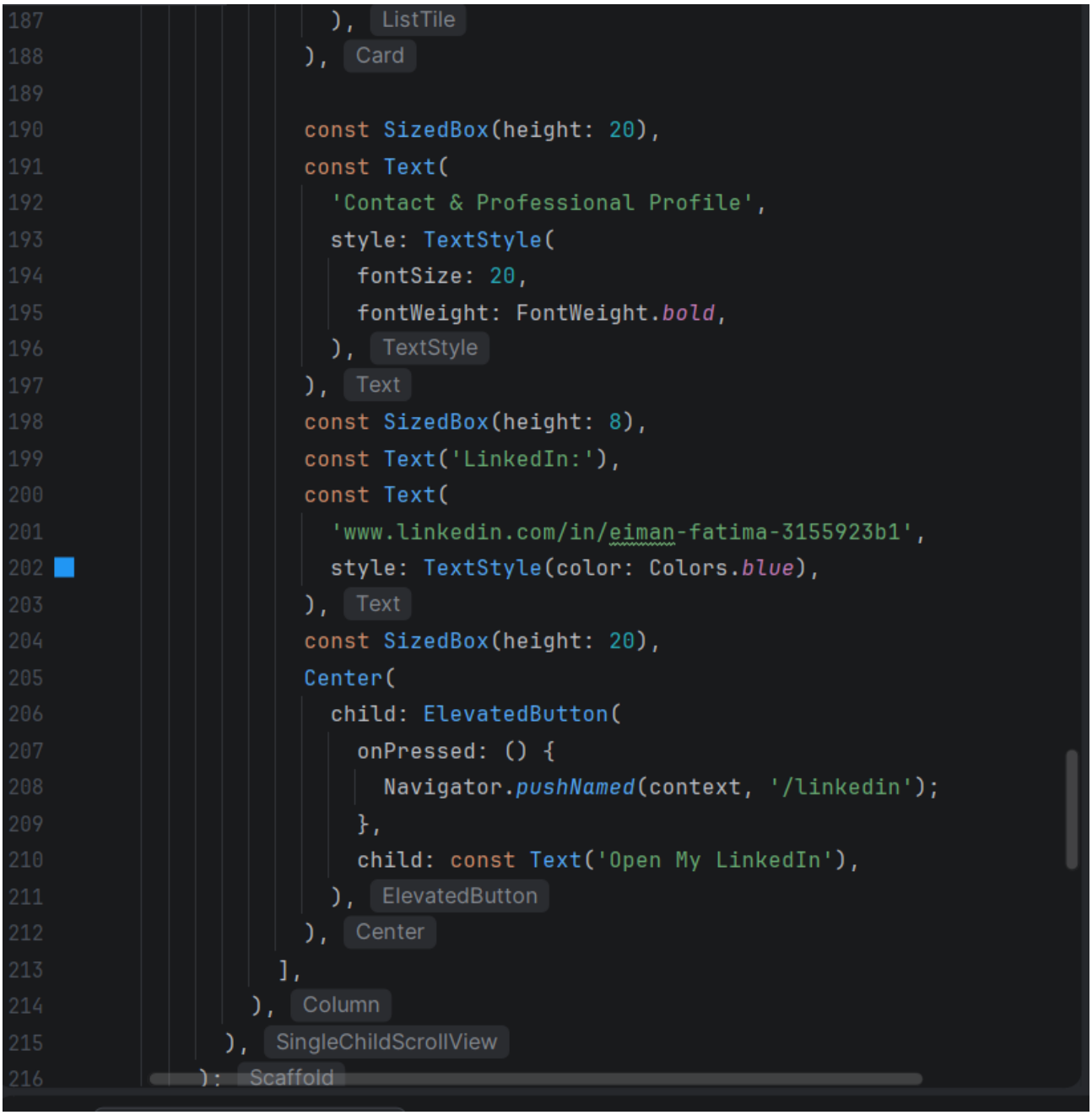
    // PORTFOLIO SCREEN
    class PortfolioScreen extends StatelessWidget {
      const PortfolioScreen({super.key});

      @override
      Widget build(BuildContext context) {
        return Scaffold(
          appBar: AppBar(
            title: const Text('About Me'),
            backgroundColor: const Color(0xFFDCEBFF),
          ), AppBar
```

```dart
body: SingleChildScrollView(
  padding: const EdgeInsets.all(16),
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Center(
        child: Hero(
          tag: 'profile',
          child: CircleAvatar(
            radius: 55,
            backgroundColor: Colors.blue,
            child: const Icon(
              Icons.person,
              size: 65,
              color: Colors.white,
            ), Icon
          ), CircleAvatar
        ), Hero
      ), Center
      const SizedBox(height: 20),
      const Center(
        child: Text(
          'EIMAN FATIMA',
          style: TextStyle(
            fontSize: 24,
            fontWeight: FontWeight.bold,
          ), TextStyle
        ), Text
      ), Center
      const SizedBox(height: 20),
```


const Text(
'Professional Summary',
style: TextStyle(
fontSize: 20,
fontWeight: FontWeight.bold,
), TextStyle
), Text
const SizedBox(height: 8),
const Text(
'Motivated Software Engineering student with a strong '
'foundation in programming, problem-solving, and soft
'development principles. Skilled in C++, Python, and
'technologies, with hands-on experience gained throug
'academic and personal projects. Eager to apply techn
'knowledge in a professional environment and further
'develop software engineering skills.',
), Text
const SizedBox(height: 20),
const Text(
'Education',
style: TextStyle(
fontSize: 20,
fontWeight: FontWeight.bold,
), TextStyle
), Text
const Card(
child: ListTile(
leading: Icon(Icons.school, color: Colors.blue),
title: Text('BS Software Engineering'),
subtitle: Text(


Resource Manager
5
'2024-2028\n'
'Air University, Kamra Campus, Pakistan\n'
'4th Semester',
), Text
), ListTile
), Card
const SizedBox(height: 20),
const Text(
'Skills',
style: TextStyle(
fontSize: 20,
fontWeight: FontWeight.bold,
), TextStyle
), Text
const Card(
child: ListTile(
leading: Icon(Icons.code, color: Colors.blue),
title: Text('C++ Programming'),
), ListTile
), Card
const Card(
child: ListTile(
leading: Icon(Icons.computer, color: Colors.blue),
title: Text('Python Programming'),
), ListTile
), Card
const Card(
child: ListTile(
leading: Icon(Icons.web, color: Colors.blue),


),
ListTile
),
Card
const SizedBox(height: 20),
const Text(
'Contact & Professional Profile',
style: TextStyle(
fontSize: 20,
fontWeight: FontWeight.bold,
),
TextStyle
),
Text
const SizedBox(height: 8),
const Text('LinkedIn:'),
const Text(
'www.linkedin.com/in/eiman-fatima-3155923b1',
style: TextStyle(color: Colors.blue),
),
Text
const SizedBox(height: 20),
Center(
child: ElevatedButton(
onPressed: () {
Navigator.pushNamed(context, '/linkedin');
},
child: const Text('Open My LinkedIn'),
),
ElevatedButton
),
Center
],
),
Column
),
SingleChildScrollView
);
Scaffold

```dart
  }
}

// LINKEDIN WEBVIEW SCREEN
class LinkedInScreen extends StatefulWidget {
  const LinkedInScreen({super.key});

  @override
  State<LinkedInScreen> createState() => _LinkedInScreenState();
}

class _LinkedInScreenState extends State<LinkedInScreen> {
  late final WebViewController controller;

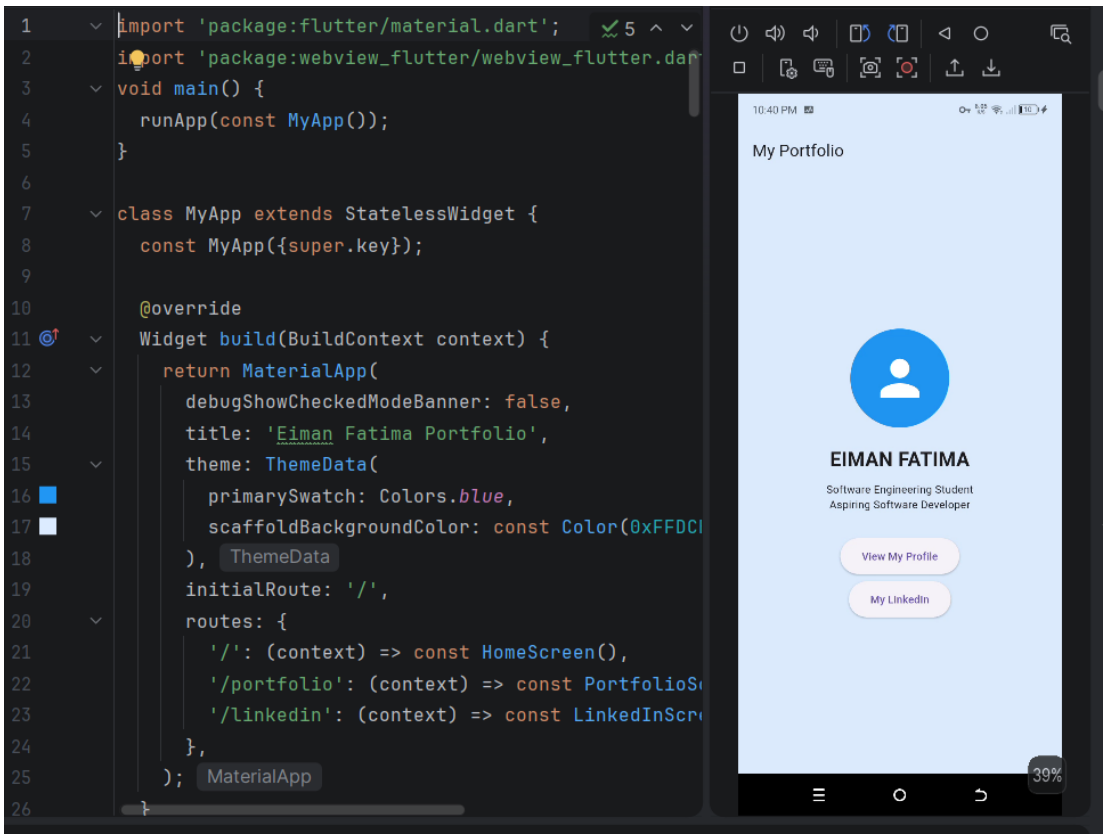
  @override
  void initState() {
    super.initState();

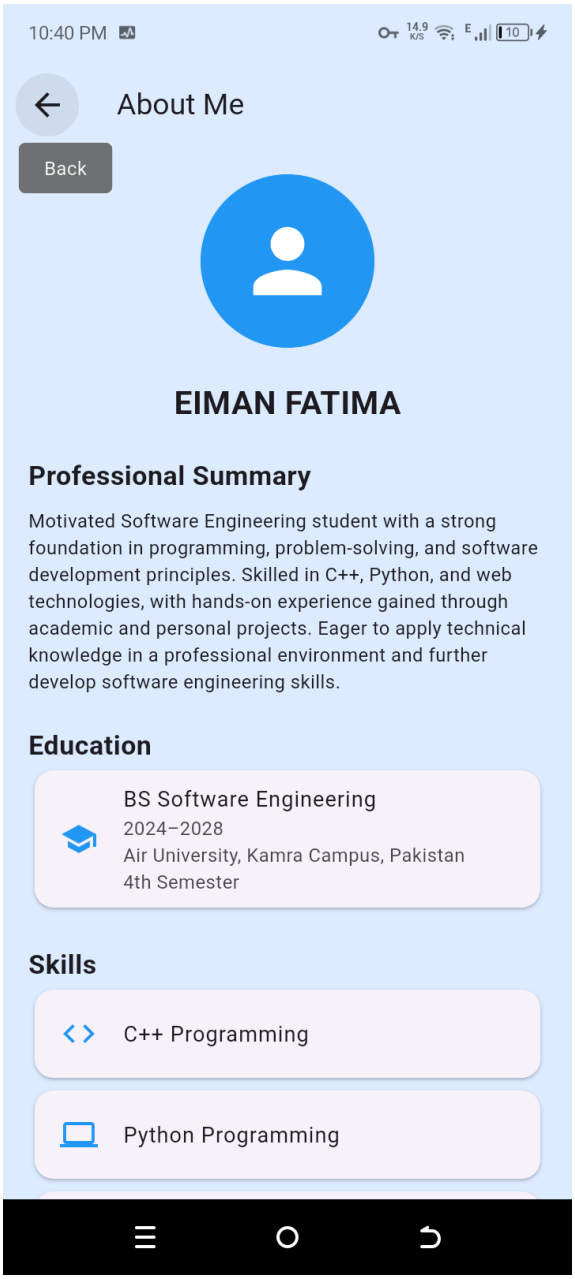
    controller = WebViewController()
      ..setJavaScriptMode(JavaScriptMode.unrestricted)
      ..loadRequest(
        Uri.parse(
          'https://www.linkedin.com/in/eiman-fatima-3155923b1',
        ),
      );
  }

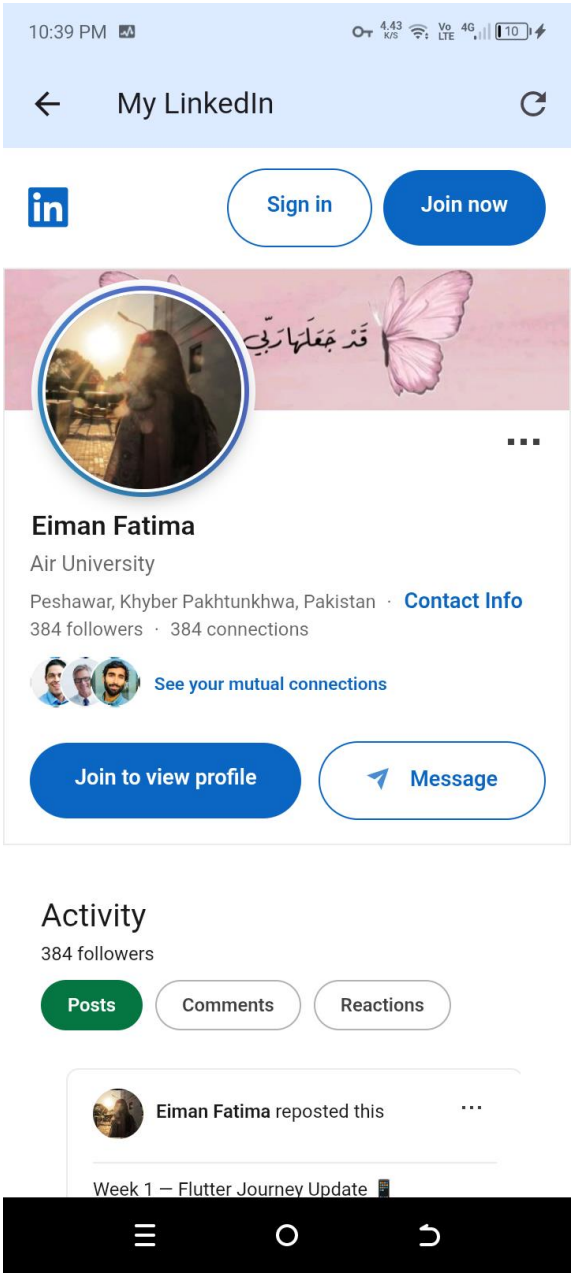
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
```

```
            title: const Text('My LinkedIn'),
            backgroundColor: const Color(0xFFDCEBFF),
            actions: [
              IconButton(
                onPressed: () {
                  controller.reload();
                },
                icon: const Icon(Icons.refresh),
              ), IconButton
            ],
          ), AppBar
          body: WebViewWidget(controller: controller),
        ); Scaffold
      }
    }
```

OUTPUT:




10:40 PM
About Me
Back
EIMAN FATIMA
Professional Summary
Motivated Software Engineering student with a strong foundation in programming, problem-solving, and software development principles. Skilled in C++, Python, and web technologies, with hands-on experience gained through academic and personal projects. Eager to apply technical knowledge in a professional environment and further develop software engineering skills.
Education
BS Software Engineering
2024–2028
Air University, Kamra Campus, Pakistan
4th Semester
Skills
C++ Programming
Python Programming


10:39 PM
My LinkedIn
Sign in
Join now
Eiman Fatima
Air University
Peshawar, Khyber Pakhtunkhwa, Pakistan · Contact Info
384 followers · 384 connections
See your mutual connections
Join to view profile
Message
Activity
384 followers
Posts
Comments
Reactions
Eiman Fatima reposted this
Week 1 — Flutter Journey Update